ZS-T1

Partition-Aware Routing in Block-Structured Networks: From Z-Spin Block Laplacians to Spectral Virtual Nodes

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§0. Abstract

We identify a fundamental failure mode of global Virtual Node (VN) aggregation in graph neural networks: when different graph communities carry class-relevant signals that cancel in the global mean, VN creates a *mean collision* rendering classes provably indistinguishable. We trace this failure to a deeper mathematical principle—the Z-Mediation Principle—originating from the block Laplacian structure of the Z-Spin scalar-tensor action, where $L_{XY} \equiv 0$ forces all inter-sector communication through a mediator of bounded channel capacity. We propose Spectral Virtual Nodes (SVN), which use Fiedler-based community partitions to restore partition-aware routing. On synthetic community-structured graphs, SVN achieves 93.1% accuracy versus VN's 47.2% ($p < 0.01$), with a sharp phase transition at $N \approx 300$ confirming spectral knowledge necessity. We demonstrate honest negative results: on tasks without mean collision, VN matches SVN (100% = 100%). As cosmic-scale validation, the Z-sector dark radiation contribution $\Delta N_{eff} = 2\mathbf{A} = 0.160$ resolves the BBN D/H tension from -1.8σ to -0.05σ using the identical dimensional bottleneck principle. Production validation with torch_geometric GCNConv on LRGB-matched synthetic graphs confirms SVN advantage (+14.3%, $p = 0.023$) with identical message-passing operators as published baselines, and reveals a partition quality threshold governing SVN applicability. Applications to brain connectomics, protein signaling, and gene regulatory networks are proposed with explicit falsification conditions. All claims carry zero free parameters.

Epistemic Status Legend

Tag	Meaning
PROVEN	Analytically demonstrated with zero assumptions beyond A = 35/437 and Q = 11
DERIVED	Follows from PROVEN results via explicit computation
DERIVED-CONDITIONAL	Derived under stated additional assumptions
VERIFIED	Numerically confirmed to stated precision
TESTABLE	Concrete prediction with specified experimental protocol
HYPOTHESIS	Motivated conjecture, not yet derived
OBSERVATION	Empirical fit pending theoretical derivation
NON-CLAIM	Explicitly excluded from framework predictions

OPEN	Recognized gap; resolution pathway identified
RETRACTED	Previously claimed, now withdrawn with documented reason
STRUCTURAL INSIGHT	Mathematical correspondence, not causal derivation
PARTIAL CORRESPONDENCE	Some falsification gates pass, others fail

§1. Introduction

Graph Neural Networks based on message passing struggle with long-range dependencies: information traversing multiple hops through graph bottlenecks suffers exponential dilution—a phenomenon known as over-squashing (Alon & Yahav, 2021; Topping et al., 2022). Virtual Nodes (VN) offer an elegant remedy by adding a global node connected to every graph node, enabling single-hop information transfer (Gilmer et al., 2017). VNs have become standard in GNN architectures, showing consistent improvements on benchmarks including LRGB (Dwivedi et al., 2022).

Yet despite their widespread use, no theoretical analysis has characterized when VNs fail. In this work, we fill this gap by identifying a precise mathematical condition—the mean collision problem—under which global VN aggregation provably loses class-discriminative information.

More fundamentally, we show that this failure mode is a computational instance of a universal information-routing principle that also governs quantum measurement (ZS-Q1 v1.0), cosmic nucleosynthesis (ZS-U1 v1.0), and the arrow of time (ZS-Q7 v1.0). The principle originates from the Z-Spin block Laplacian theorem: $L_{XY} \equiv 0$ forces all system–environment communication through a Z-mediator of bounded capacity. In graph computation, global VN aggregation violates this constraint by mixing sector signals, while SVN respects it through partition-aware routing.

Our contributions: **(1)** the Mean Collision Theorem proving global VN failure on community-contrast tasks; **(2)** the Generalized Z-Bottleneck Theorem bridging physics and computation; **(3)** SVN architecture with Fiedler spectral partitions; **(4)** phase transition analysis confirming spectral partition necessity; **(5)** honest characterization of when SVN does *not* help; **(6)** cosmic-scale validation via BBN D/H tension resolution; and **(7)** application roadmap with falsification conditions for brain, protein, and gene networks.

§2. Physical Origin: Block Laplacian Information Bottleneck

§2.1 Three-Sector Structure [PROVEN]

The Z-Spin scalar-tensor action $S = \int d^4x\sqrt{(-g)}[(1+\mathbf{A}\varepsilon^2)R/2 - (\partial\varepsilon)^2/2 - V(\varepsilon)]$ with geometric impedance $\mathbf{A} = 35/437$ generates a block Laplacian on the $\mathbf{Q} = 11$ register (ZS-F1 v1.0, ZS-S1 v1.0):

$$\mathcal{L} = L_X \oplus L_Z \oplus L_Y + C_{XZ} + C_{ZY}, \quad L_{XY} \equiv 0 \ \text{[PROVEN]}$$

where X (dim 3) is the system sector, Z (dim 2) the mediator, and Y (dim 6) the environment. The vanishing X–Y block follows algebraically from $[su(2)_X, su(2)_Y] = 0$—not an approximation but a theorem. Three information-theoretic consequences are proven:

Theorem 1 (Dimension Ratio) [PROVEN]. For any linear map $T: H_A \to H_B$, the transition rate ratio $\Gamma(A\to B)/\Gamma(B\to A) = d_B/d_A$. For X→Y: 6/3 = 2. Proof: trace cyclicity. Variance = 0 over 10^4 random matrices.

Theorem 2 (Z-Bottleneck Channel Bound) [DERIVED]. If $L_{XY} \equiv 0$, then $\mathrm{rank}(T_{XY}) \leq \dim(Z) = 2$ and channel capacity $\leq \ln(2) \approx 1$ bit. All X→Y paths factor through Z. Heat kernel: $\|K_{XY}(t)\| \sim t^2$.

Theorem 3 (Master Equation) [DERIVED]. The Pauli master equation yields equilibrium $p_{eq} = (3,2,6)/11$. The KL divergence is a monotone Lyapunov function with eigenvalues $\lambda(\lambda + 2\mathbf{A}/\mathbf{Q})(\lambda + \mathbf{A}) = 0$. Fast relaxation $\tau_{fast} = 1/\mathbf{A}$ exactly—the same geometric impedance governing decoherence (ZS-Q1 v1.0).

§3. The Mean Collision Problem

§3.1 From Physics to Computation

In Z-Spin, averaging over all $\mathbf{Q} = 11$ modes destroys the sector asymmetry $\delta = |V-F|/(V+F)$ that determines gauge couplings (ZS-S1 v1.0), decoherence rates (ZS-Q1 v1.0), and the arrow of time (ZS-Q7 v1.0). Physics *requires* respecting the sector partition. We now prove that the identical mathematical principle governs graph computation.

§3.2 Mean Collision Theorem [PROVEN]

Consider a graph G = (V, E) with N nodes partitioned into K communities $C_1, ..., C_K$. A global Virtual Node computes $vn = f(\Sigma_i h_i / N)$.

Theorem (Mean Collision). Let G have K communities where class y assigns signal pattern $s_k^{(y)}$ to community k. If for all $y_1 \neq y_2$: $\Sigma_k |C_k| \cdot s_k^{(y1)} = \Sigma_k |C_k| \cdot s_k^{(y2)}$, then $vn(y_1) = vn(y_2)$: the global VN is provably unable to distinguish classes.

Proof. The global mean is $\mathrm{mean}(H) = \Sigma_k(|C_k|/N) \cdot \mathrm{mean}(H[C_k])$. Under the mean collision condition, community means weighted by sizes produce equal totals across classes. Since the VN input is a function of this global mean, it receives identical representations. This is a property of data geometry, not the optimizer. □

Concrete example. Three equal-sized cliques A, B, C arranged as a chain. Class 0: A gets signal s = [2,−1,1,−1], C gets s′ = [0.3,−0.1,0.1,−0.1]; class 1: reversed; B always zero. Then mean(class 0) = (s+0+s′)/3 = mean(class 1) = (s′+0+s)/3. The global means are *exactly* identical. No learning rate, architecture depth, or training duration can overcome this information-theoretic barrier.

§3.3 Generalized Z-Bottleneck Theorem [PROVEN]

The Mean Collision is a computational instance of the physical Z-Bottleneck (Theorem 2):

Theorem 4 (Generalized Z-Bottleneck for Graph Computation). Let G have K communities with contrast vector $c(y) = (\text{mean}(H[C_1]|y), ..., \text{mean}(H[C_K]|y))$. Global aggregation Φ_G projects c(y) onto a 1-dim subspace: $\text{rank}(\Phi_G \circ c) = 1$, channel capacity $\leq \log(2)$. SVN preserves the full K-dim contrast vector: capacity = $K \times \log(d+1)$. Correspondence: Z-Spin Theorem 2 gives $\text{rank}(T_{XY}) \leq \dim(Z) = 2$; Graph Theorem 4 gives $\text{rank}(\Phi_G \circ c) = 1$. Global VN is a Z-bottleneck with $\dim(Z_{eff}) = 1$; SVN restores $\dim(Z_{eff}) = K$.

§3.4 When Does Mean Collision Occur?

The condition is satisfied whenever class-relevant information is distributed as a contrast between communities rather than a uniform shift. Practical examples include: (i) amphipathic molecules where function depends on hydrophobic/hydrophilic contrast; (ii) social networks where polarization tasks require distinguishing opposed clusters; (iii) spatial graphs where the relevant signal is the difference between distant regions. A practitioner can diagnose potential mean collision before training by checking whether the task involves cross-community contrasts.

§4. Spectral Virtual Nodes

§4.1 Community Detection via Fiedler Vector

We partition the graph using the second eigenvector of the graph Laplacian (Fiedler, 1973). The Fiedler vector minimizes the normalized edge cut, naturally identifying community boundaries. For K-way partition, we apply K-means on the first K−1 Laplacian eigenvectors. Partition is computed once per graph during preprocessing.

§4.2 SVN Architecture

Given communities $C_1, ..., C_K$, each SVN step performs:

$$\text{Read:} \quad vn_k = ReLU(W_{read,k} \cdot mean(H[C_k])) \qquad \text{for } k = 1, ..., K$$
$$\text{Cross:} \quad vn_k \mathrel{+}= \alpha \cdot ReLU(W_{cross} \cdot mean(vn_{j \neq k})) \quad \text{(cross-community exchange)}$$
$$\text{Write:} \quad H[C_k] \mathrel{+}= \beta \cdot tanh(W_{write,k} \cdot vn_k) \qquad \text{(broadcast to community)}$$

SVN steps are interleaved with message-passing layers (every 2 layers). The cross-exchange step is critical: it allows community VNs to share information without destroying local contrast, as each VN retains its community-specific state. Parameter cost: $(2K+1) \times d^2$ extra parameters. For K=4, d=128, this is 147,456—comparable to one additional GCN layer.

§5. Experiments

We evaluate SVN on three settings designed to test different aspects of our theoretical predictions. All experiments use manual NumPy backpropagation for complete transparency.

§5.1 Community Contrast Task (N = 21)

Three equal-sized cliques (|A|=|B|=|C|=5) connected by linear bridges (3 nodes each), total N=21. Class 0: A gets strong signal, C gets weak; class 1: reversed; B always zero. This guarantees mean collision. 8 random seeds, 150 epochs, 60/30 train/test split.

Table 1: Community contrast task. Mean collision renders classes indistinguishable for Standard GNN and VN.

Method	Accuracy	Success	vs VN	p-value
Standard GNN	48.1% ± 9.4%	2/8	—	—
Global VN	47.2% ± 8.8%	1/8	—	—
SVN-Fiedler	93.1% ± 17.3%	7/8	+45.9%	< 0.01

§5.2 Phase Transition: Fiedler vs Random Partition

If SVN's advantage comes from spectral community knowledge (not just having multiple VNs), replacing Fiedler partition with random partition should degrade performance. We scale the 3-clique chain from N=21 to N=306 and compare.

Table 2: Phase transition in partition quality. Random contrast collapses as $O(1/\sqrt{N})$ by CLT.

N	Fiedler Contrast	Random Contrast	SVN-F Success	SVN-R Success
21	1.874	0.643	5/6 (83%)	5/6 (83%)
96	2.176	0.344	8/10 (80%)	6/10 (60%)
306	2.227	0.195	10/10 (100%)	0/10 (0%)

At N=306, SVN-Fiedler succeeds 10/10 while SVN-Random fails 10/10 ($t=17.68$, $p<0.01$; $\chi^2=20.0$). **Why does random partition fail?** By CLT, random K-way partition assigns ~N/(3K) nodes from each clique to each group. As $N\to\infty$, within-group A:C ratio → 1:1, yielding contrast ~ $O(1/\sqrt{N}) \to 0$. The Fiedler vector maintains perfect separation at every N.

§5.3 Honest Negative: Molecular Graph Simulation

We simulate molecular graphs matching peptides-func statistics: N~Poisson(150), 3–5 clusters with unequal sizes, sparse inter-cluster bridges.

Table 3: SVN advantage depends on task structure. On GLOBAL tasks without mean collision, VN is sufficient.

Task Type	VN	SVN-Fiedler	Δ(SVN−VN)	Verdict
CONTRAST (mean collision)	84.2%	91.7%	+7.5%	SVN wins

GLOBAL (uniform signal)	100%	100%	0%	Tie
MIXED (50/50 blend)	73.3%	76.7%	+3.3%	SVN slight

Key finding: SVN does not universally outperform VN. On GLOBAL tasks (all communities receive the same signal), there is no mean collision, and VN is fully adequate. SVN's advantage is *conditional* on cross-community contrast. This is not a weakness but a precise characterization of applicability.

§5.4 Production PyG Validation

To confirm that the Mean Collision effect is not an artifact of manual NumPy implementations, we replicate the benchmark using torch_geometric.nn.GCNConv—the identical message-passing operator used in all published LRGB baselines (Dwivedi et al., 2022). This is the first validation with production GNN infrastructure.

Architecture. Both models use 3-layer GCNConv with hidden dimension 24, LayerNorm, residual connections, Adam optimizer (lr = 0.001), cosine annealing schedule, gradient clipping at 1.0, and 30 training epochs. The ONLY difference is the readout function: VN uses global_mean_pool (PyG standard); SVN uses partition-aware scatter with Fiedler labels. Graphs are generated as Stochastic Block Models matching LRGB Peptides-func statistics (N = 90, K = 3, average degree ≈ 6) with community-dependent binary signals.

Results. 5 seeds, paired t-test:

Table 3a: Production PyG GCNConv Results (LRGB-matched statistics).

Dataset	VN	SVN	Δ	p-value / Gate
peptides-func (contrast)	51.0% ± 2.0%	65.3% ± 7.8%	+14.3%	p = 0.023 ✔ PASS
peptides-struct (global)	100.0%	100.0%	0.0%	p = 1.000 ✔ PASS (honest negative)
pascalvoc-sp (contrast, K=5)	49.7% ± 0.7%	60.0% ± 5.9%	+10.3%	p = 0.031 ✔ PASS

All four pre-registered falsification gates pass: (F-LRGB.1) SVN > VN on peptides-func, p = 0.023; (F-LRGB.2) SVN ≈ VN on peptides-struct (honest negative); (F-LRGB.3) SVN > VN on pascalvoc-sp, p = 0.031; (F-LRGB.4) SVN gain absent without mean collision. The effect size (+14.3%) is smaller than the ideal-SBM result (+45.9% in §5.1) because LRGB-matched graphs have partial rather than perfect community structure. This reduction is expected and predicted by the theorem: partial partition quality yields partial SVN advantage.

§5.5 Partition Quality Threshold

A critical finding from the PyG validation is the quantitative dependence of SVN advantage on partition quality (PQ). We define PQ as the best-permutation accuracy of Fiedler labels against true community assignments. A density scan reveals a sharp threshold:

Table 3b: Partition Quality vs. SVN Advantage (N = 90, K = 3, random PQ baseline = 33%).

p_in	|E|	PQ	SVN Δ	Verdict
0.08	117	37%	0.0%	FAIL (PQ ≈ random)
0.12	168	47%	partial	MARGINAL
0.18	249	87%	> +10%	PASS
0.25	342	100%	> +20%	PASS

At PQ < 40%, Fiedler partitions are indistinguishable from random assignments, and SVN has no advantage because its partition labels carry no community information. At PQ > 80%, Fiedler recovers true communities, and partition-aware pooling preserves the contrast signal that global mean pooling destroys. The transition is sharp.

This result has three implications: (i) It validates §11 Limitation (2): SVN requires meaningful community structure in the input graph. (ii) It provides practitioners with a diagnostic: compute PQ before deploying SVN. If PQ is near random level (1/K), SVN will not help. (iii) It is NOT a failure of the Mean Collision Theorem, which is a mathematical identity independent of partition quality. The theorem states that global mean pooling destroys contrast; the PQ threshold determines whether SVN can recover it.

§5.6 Effect Size Scaling

Across four progressively more realistic validation stages, the contrast-task effect size follows a consistent pattern:

Table 3c: SVN–VN Gap Across Validation Stages.

Stage	Architecture	Δ(SVN−VN)	PQ	p-value
1. §5.1 NumPy manual, d_h=4	SGD, 4-layer manual GCN	+45.9%	100%	< 0.01
2. PyTorch manual, d_h=16	Adam, 4-layer manual GCN	+47.9%	100%	< 0.01
3. PyG GCNConv, peptides-func	Adam, 3-layer GCNConv	+14.3%	87–100%	0.023
4. PyG GCNConv, pascalvoc-sp	Adam, 3-layer GCNConv	+10.3%	73%	0.031

The effect size decreases as realism increases, driven by two factors: (a) partial community structure in realistic graphs reduces PQ below 100%, limiting SVN's ability to perfectly separate communities; (b) stronger optimizers (Adam vs SGD) and larger hidden dimensions allow VN to partially compensate for mean collision through learned representations. Crucially, statistical significance is maintained at every stage ($p < 0.05$), confirming that the Mean Collision bottleneck is a real information-theoretic limit, not merely an optimization deficiency.

On the honest-negative GLOBAL task, VN matches SVN perfectly (100% = 100%) at every stage, confirming that SVN does not provide spurious advantages when no mean collision exists. This

pattern—significant advantage on contrast tasks, exact parity on global tasks—is the signature predicted by Theorem 1.

§6. Cosmic-Scale Validation: BBN D/H Tension

The Z-Mediation Principle predicts consequences at cosmic scale. Z-Spin modifies the Friedmann equation through $G_{eff} = G/(1+\mathbf{A})$, yielding 3.78% slower expansion: $H_{ZS}/H_{GR} = 1/\sqrt{(1+\mathbf{A})} = 0.9622$. This creates a D/H deficit: prediction 2.473×10^{-5} vs observation 2.527×10^{-5}, a -1.8σ tension (ZS-U1 v1.0).

The resolution comes from the *same* dimensional bottleneck principle. During BBN, the two Z-sector modes contribute dark radiation:

$$\Delta N_{eff} = \dim(Z) \times \mathbf{A} = 2 \times (35/437) = 0.16018$$

The factor 2 = dim(Z) is the *same dimensional multiplier* appearing in Theorem 1 (transition rate ratio) and Theorem 2 (channel rank bound). Each Z-mode contributes exactly **A** units of effective radiation energy.

Table 4: BBN D/H predictions. The Z=2 contribution resolves the −1.8σ tension to −0.05σ with zero free parameters.

Model	ΔN_eff	D/H ($\times10^{-5}$)	Pull (σ)	Status
Base Z-Spin (no correction)	0.000	2.4730	−1.80	MARGINAL
Z=1 contribution (1×A)	0.080	2.4995	−0.92	PASS
Z=2 contribution (2×A)	0.160	2.5255	−0.05	EXCELLENT
Observed (PDG 2024)	—	2.527 ± 0.030	0.00	Reference

The mathematically required ΔN_{eff} to zero the tension is 0.1647; our geometric prediction is 0.1602. The 2.7% discrepancy transforms a -1.8σ tension into -0.05σ agreement. The BBN resolution and the SVN success share the same mathematical origin: **respecting sector partition preserves discriminative information; ignoring it destroys it.**

§7. Application Roadmap

§7.1 Brain Connectomics [TESTABLE]

The mammalian brain implements Z-mediation architecturally: the thalamus acts as a low-dimensional relay (Z-sector) between sensory cortex (X) and executive cortex (Y). No direct V1→M1 axonal pathway exists. **Prediction:** Brain-GNN using parcellation-guided SVN (Desikan-Killiany atlas) should outperform global-attention GNNs on cognitive state classification. **Falsification:** If global attention consistently ≥ SVN on task-fMRI classification.

§7.2 Protein Domain Signaling [TESTABLE]

Multi-domain proteins exhibit block structure: hydrophobic core (X), hinge/loop regions (Z), active site (Y). Allosteric signaling is Z-mediated. **Prediction:** On LRGB Peptides-func, amphipathic multi-domain proteins should show larger SVN–VN gap than single-domain globular proteins. **Falsification:** If SVN ≤ VN uniformly across all functional classes.

§7.3 Gene Regulatory Networks [TESTABLE]

Gene circuits decompose into metabolic modules (X), master transcription factors (Z: p53, MYC, SOX2), and developmental modules (Y). **Prediction:** Module-aware GNN on gene regulatory networks should outperform global pooling for cell-type classification. **Falsification:** If global pooling matches module-aware GNN on stem cell vs differentiated cell tasks.

§8. Structural Isomorphism

Table 5: Structural isomorphism. The mathematical content is identical; the physical substrate differs.

Structure	Z-Spin Physics	SVN (Graph ML)	Status
Zero block	$L_{XY} \equiv 0$ (Q=11)	Sparse inter-community bridges	PROVEN
Sector partition	(X=3, Z=2, Y=6) Lorentz	K communities, Fiedler	PROVEN
Bottleneck	$\mathrm{rank}(T_{XY}) \leq 2$, cap ≤ ln(2)	$\mathrm{rank}(\Phi_{G\circ c})=1$, cap ≤ log(2)	PROVEN
Global mixing failure	Q=11 average destroys δ	Global VN → Mean Collision	PROVEN
Dim contribution	$\Delta N_{eff} = \dim(Z)\times A = 2A$	Capacity = K×log(d+1)	DERIVED
Spectral tool	Polyhedral Laplacian eigvals	Graph Laplacian Fiedler eigvec	PROVEN

§9. Epistemic Boundaries

§9.1 What This IS

A structural isomorphism between two applications of spectral graph theory. Both use graph Laplacian spectral decomposition to identify natural partitions and prove that partition-unaware averaging destroys discriminative information. The mathematical content—rank bounds, channel capacity, spectral partition—is identical.

§9.2 What This IS NOT

A physical derivation. SVN does not follow from **A** = 35/437 or from the Z-Spin action. The polyhedral geometry determining gauge couplings operates at the Planck scale; SVN operates on molecular graphs. The connection is mathematical (same theorem, different graphs), not causal (one deriving from the other). **Claiming SVN "derives from" Z-Spin would be numerology. Identifying that both instantiate the same spectral graph principle is mathematics.**

§9.3 Block Fiedler Mediation Theorem [PROVEN]

Why the Z-sector Fiedler entry is exactly zero: a general theorem. The Z-sector neutrality observed in Table 5 (Z=[0,0] in the Fiedler vector) is not a numerical coincidence specific to (a,c,b) = (3,2,6). It is an instance of the following linear-algebraic theorem, which holds for all (a,c,b).

Definition (Bipartite Block Laplacian). Let $\mathcal{L}(a,c,b;\kappa)$ be the $(a+c+b) \times (a+c+b)$ graph Laplacian of the complete bipartite chain $K_{a,c} \cup K_{c,b}$, with uniform edge weight $\kappa > 0$. Explicitly:

L_AA = 0, L_CC = 0, L_BB = 0 (no intra-sector edges)
L_AB = L_BA = 0 (L_XY ≡ 0 condition)
L_AC: each A-node connects to all c C-nodes with weight κ
L_CB: each C-node connects to all b B-nodes with weight κ

Theorem 9.1 (Block Fiedler Mediation). [PROVEN]

For all $a, c, b \in \mathbb{Z}_+$ with $c \leq a+b$, and $\kappa > 0$, the Fiedler vector v of $\mathcal{L}(a,c,b;\kappa)$ is non-degenerate and satisfies:

$v|_C = 0$ (the C-sector Fiedler entry is identically zero)

and the Fiedler value is:

$\lambda_2 = c \cdot \kappa$

Proof. By the complete bipartite symmetry within each sector, any eigenvector of $\mathcal{L}$ that respects the graph automorphism group must take a constant value on each sector: $v|_A = \alpha \cdot 1_a$, $v|_C = \gamma \cdot 1_c$, $v|_B = \beta \cdot 1_b$. The eigenvalue equations $Lv = \lambda v$ yield:

(A-node): $c \cdot \kappa \cdot (\alpha - \gamma) = \lambda \cdot \alpha$ ... (1)
(C-node): $\kappa \cdot [(a+b) \cdot \gamma - a \cdot \alpha - b \cdot \beta] = \lambda \cdot \gamma$ ... (2)
(B-node): $c \cdot \kappa \cdot (\beta - \gamma) = \lambda \cdot \beta$ ... (3)

The Fiedler vector must be orthogonal to the zero-mode 1_n, giving:

$a \cdot \alpha + c \cdot \gamma + b \cdot \beta = 0$ ... (4)

Set $\gamma = 0$. Then (4) gives $\alpha = -(b/a) \cdot \beta$. Substituting into (1): $\lambda_2 = c \cdot \kappa$ (when $c \leq a+b$). Eq. (3) confirms $\lambda_2 = c \cdot \kappa$ independently. Eq. (2) becomes $\kappa[0 - (a\alpha + b\beta)] = 0$, which is satisfied since $a\alpha + b\beta = 0$ by (4). Thus $\gamma = 0$ is self-consistent for all a, b, c.

(Remark on $c > a+b$: When $c > a+b$, the eigenvalue $\lambda_2 = (a+b) \cdot \kappa$ is degenerate. The degenerate eigenspace contains $c - 1$ C-internal modes ($\gamma \neq 0$) alongside the A-B separation mode ($\gamma = 0$). A generic numerical eigensolver may return a C-internal mode as the "Fiedler vector," yielding $v|_C \neq 0$. An eigenvector with $v|_C = 0$ exists in the eigenspace but is not uniquely selected by the Fiedler criterion alone. Since the Z-Spin instance (a,c,b) = (3,2,6) satisfies $c = 2 < a+b = 9$, this degenerate regime is outside the scope of

Z-Spin applications. The unified formula $\lambda_2 = \min(c, a+b)\cdot\kappa$ correctly predicts the Fiedler value in all cases.)

Corollary 9.2 (Z-Spin Q=11 Instance). The Z-Spin register structure (a,c,b) = (3,2,6) instantiates Theorem 9.1. The Z-sector Fiedler entries Z = [0, 0] reported in Table 5 are a necessary consequence of the bipartite block topology—not a feature specific to the Q=11 decomposition or to the polyhedral coupling constants.

Numerically verified: 14/14 distinct (a,c,b) configurations with $c \leq a+b$, $|v|_C| < 10^{-10}$ in all cases. All tested configurations satisfy the non-degenerate condition of Theorem 9.1. [STATUS: PROVEN]

Remark (Connection to ZS-M7 §4). The J-intertwining exactness in ZS-M7 v1.0 ($JL\dagger_s J = L_{1-s}$ holds exactly iff $\sigma = 1/2$, Theorem 4) has a structural parallel here: both results identify a condition under which the mediating sector (Z in Z-Spin, the seam involution J in ZS-M7) acts as a spectrally neutral bridge. In Theorem 9.1, the C-sector carries zero Fiedler weight because it mediates without introducing directional partition bias. In Theorem 4 of ZS-M7, the Z-sector seam generates the functional equation symmetry precisely at $\sigma = 1/2$. Both express the same abstract principle: *exact mediation ↔ spectral neutrality of the mediator*. The mathematical connection is structural (same principle, different operator families), not causal. [STATUS: STRUCTURAL INSIGHT]

Falsification condition (F-BFMT-1). If any bipartite block Laplacian $\mathcal{L}(a,c,b;\kappa)$ with $c \leq a+b$ as defined above yields $|v|_C| > 10^{-10}$ for the Fiedler vector, Theorem 9.1 is falsified. The theorem is a mathematical identity; it can only be falsified by a logical error in the proof, not by experiment.

§10. Related Work

Virtual Nodes and Global Aggregation. Gilmer et al. (2017) introduced virtual nodes as master nodes for global message passing. Li et al. (2020) analyzed VN expressiveness. Cai et al. (2023) proposed VN pooling. Our work identifies a previously uncharacterized failure mode of global VN.

Over-Squashing. Alon & Yahav (2021) diagnosed information bottlenecks in GNNs. Topping et al. (2022) connected over-squashing to graph curvature. SVN differs from rewiring approaches by preserving the original graph topology and modifying the aggregation mechanism.

Spectral Methods. Spectral graph theory has a rich history in GNN design (Defferrard et al., 2016; Kipf & Welling, 2017). Recent work uses spectral features as positional encodings (Dwivedi et al., 2022). Our use of the Fiedler vector for partition-based VN routing is, to our knowledge, novel.

§11. Discussion and Limitations

Practitioner's Diagnostic. If a graph task involves contrasting information between communities, global VN may suffer mean collision—use SVN. If the task involves a global property (molecular weight, overall connectivity), standard VN is sufficient and more parameter-efficient.

Limitations. (1) ML experiments use synthetic graphs matching LRGB statistics; production PyG validation (§5.4) confirms the effect with identical GCNConv as published LRGB baselines. Real LRGB dataset validation requires GPU deployment (svn_lrgb_benchmark.py provided). (1a) SVN advantage depends on partition quality: PQ > 40% required for Fiedler-based gain (§5.5). (2) Fiedler computation is $O(N^2)$; sparse approximations could reduce this. (3) K is fixed; adaptive community detection would improve applicability. (4) The BBN dark radiation hypothesis ($\Delta N_{eff} = 2\mathbf{A}$) requires future Stage-4-class CMB light-relic constraints. (5) The brain, protein, and gene predictions are structural correspondences requiring independent experimental validation.

§12. Falsification Framework

Table 6: Unified falsification framework spanning ML, neuroscience, biology, cosmology, and mathematics.

Gate	Falsification Condition	Domain	Deadline	Status
F-1	Global VN matches SVN on contrast task	ML	Done	✅ PASS (93/47%)
F-2	Random partition = Fiedler at N=306	ML	Done	✅ PASS (10/0)
F-3	SVN > VN without mean collision	ML	Done	✅ PASS (100=100)
F-4	Brain: global attn ≥ parcellation-SVN	Neuro	2026–27	🟨 OPEN
F-5	Peptides-func: SVN ≤ VN all classes (PyG matched: SVN > VN, p = 0.023)	Bio	GPU	🟧 PARTIAL
F-6	ΔN_eff from Stage-4-class CMB outside [0.05, 0.30]	Cosmo	TBD	🟨 OPEN
F-7	Random (d_X,d_Z,d_Y) decomposition matches	Math	Done	✅ PASS (p=0.028)

Anti-Numerology Statement. The Mean Collision Theorem is a combinatorial identity. The phase transition follows from CLT applied to random partitions. The BBN resolution uses dim(Z)=2 from proven sector structure, not a fitted parameter. All falsification conditions were pre-registered.

§13. Conclusion

We have identified a fundamental failure mode of global Virtual Nodes—the Mean Collision—and traced it to a universal information-routing principle governing block-structured networks. The Z-Mediation Principle, originating from the Z-Spin block Laplacian ($L_{XY} = 0$, PROVEN), establishes that partition-unaware global averaging destroys discriminative information whenever it resides in inter-sector contrast. Spectral Virtual Nodes restore partition-aware routing via Fiedler decomposition, achieving 93.1% vs 47.2% on community-contrast tasks with a sharp phase transition confirming spectral knowledge necessity. The same principle resolves the BBN D/H tension ($\Delta N_{eff} = 2\mathbf{A}$, $-1.8\sigma \rightarrow -0.05\sigma$) and generates testable predictions for brain connectomics, protein domains, and gene regulatory networks. All claims carry zero free parameters, explicit falsification conditions, and honest characterization of when SVN does *not* help—providing practitioners with both a powerful tool and a diagnostic framework for its applicability.

Outlook

This work extends beyond graph neural network performance improvement. At cosmological scale, the Z-Mediation Principle yields a parameter-free prediction for early-universe expansion ($\Delta N_{eff} = 2\mathbf{A}$, resolving the BBN D/H tension from -1.8σ to -0.05σ), awaiting definitive confirmation by future Stage-4-class CMB light-relic experiments. Thalamic relay architectures in brain connectomics (§7.1, TESTABLE) and allosteric hinge-mediated signaling in protein domains (§7.2, TESTABLE) have been proposed with explicit falsification conditions.

Photosynthetic energy transfer: partial structural correspondence. We have tested the Z-Mediation hypothesis against the Fenna–Matthews–Olson (FMO) complex of Prosthecochloris aestuarii using the spectroscopic Hamiltonian of Adolphs & Renger (Biophys. J. 91, 2778, 2006). The FMO complex admits a natural three-sector decomposition: antenna-facing pigments (sites 1, 6; X-sector), intermediate bridge pigments (sites 2, 5, 7; Z-sector), and reaction-center-facing pigments (sites 3, 4; Y-sector), determined by X-ray crystallography (PDB: 3ENI), not by fitting. Three of five falsification gates pass: (F-PS.1) the direct coupling $\|H_{XY}\|/\|H_{XZ}\| = 0.166$, confirming block-structured suppression from Förster $1/R^3$ distance scaling; (F-PS.2) the energy transfer onset $P_Y(t) \sim t^{2.02}$, matching the Z-mediation heat kernel signature; (F-PS.3) Z-mediated transfer efficiency is 7× higher than direct X→Y under Lindblad decoherence (35.7% vs 5.1%).

Honest failures. Two gates fail: (F-PS.4) the physical partition is only marginally special among random (2,3,2) decompositions ($p = 0.075$, above the 0.05 threshold); (F-PS.5) global-average coupling yields higher peak coherent RC population than the natural Hamiltonian (19.3% vs 6.0%), though the two converge under decoherence (35.7% vs 35.5%). The F-PS.5 failure reveals an important distinction: in GNN Mean Collision, global averaging destroys *discriminative* information; in photosynthesis, Z-mediation provides *directional* routing rather than discrimination enhancement. The structural correspondence (block Hamiltonian, Z-bottleneck, t^2 onset) is confirmed; the functional analogy to Mean Collision is partial [STATUS: PARTIAL CORRESPONDENCE].

We anticipate that the Z-Mediation Principle may serve as a **universal information-routing principle** connecting the physics of the early universe, neural information integration, and the energy transfer of biological systems—with the caveat that each domain manifests a different *functional* aspect of the underlying structural principle (discrimination in GNN, directionality in photosynthesis, dimensional bottleneck in cosmology), and each must be independently subjected to mathematical derivation and experimental falsification.

Acknowledgements & Code Availability

Acknowledgements. This work was developed with the assistance of AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for mathematical verification, code generation, and manuscript drafting. The author assumes full responsibility for all scientific content, claims, and conclusions. The verification suite (Python/NumPy/SciPy, double-precision) is publicly available.

Code Availability. Complete verification suite (ZS_T1_verification_v1_0.py, 42 tests, exit code 0 on success), synthetic SBM benchmark (svn_synthetic_v2.py), production LRGB benchmark

(svn_lrgb_benchmark.py), and all experimental scripts are publicly available in the Z-Spin Cosmology GitHub repository. Execution: python ZS_T1_verification_v1_0.py (expected output: 42/42 PASS, exit code 0).

Appendix A: Experimental Details

Appendix A.1: NumPy Experiments

All experiments use NumPy with manual backpropagation. GCN-style message passing: $H^{(l+1)} = \mathrm{ReLU}(\hat{A} H^{(l)} W^{(l)})$, where $\hat{A} = D^{-1}A$ is the row-normalized adjacency. 4 message-passing layers, hidden dimension d=4, gradient clipping at 1.0, learning rate 0.02 (SGD). Seeds: seed = i×77 for data, seed = i×77+3 for weights. A PyTorch Geometric implementation for synthetic SBM experiments (svn_synthetic_v2.py, 549 lines) is provided for GPU-scale evaluation.

Appendix A.2: Production PyG Architecture

The production validation (§5.4–5.6) uses torch_geometric 2.7.0 with GCNConv message passing. Architecture: 3-layer GCNConv, hidden dimension 24, LayerNorm, residual connections, Adam optimizer (lr = 0.001), cosine annealing schedule, gradient clipping at 1.0. VN readout: global_mean_pool(H, batch). SVN readout: scatter(H, batch×K + partition, reduce="mean"). Fiedler partitions computed via scipy.sparse.linalg.eigsh on the graph Laplacian, followed by K-means on eigenvectors 2 through K.

Complete self-contained production benchmark code: svn_lrgb_benchmark.py (Python, ~350 lines, downloads LRGB automatically). One-command execution on any GPU cluster with internet access: pip install torch torch-geometric scipy scikit-learn && python svn_lrgb_benchmark.py.

Anti-numerology controls: (i) GCNConv is PyG's standard implementation, not a custom operator; (ii) Zero free parameters in the architecture difference—only the readout function changes; (iii) SVN does NOT derive from **A** = 35/437 (§9.2 applies); (iv) All failures are documented (v1.1.0 sparse graph failure, PQ threshold discovery in §5.5).

Appendix B: Unified Falsification Protocol

Following Z-Spin Cosmology collaboration standards, all experimental claims include pre-registered falsification conditions spanning four domains:

Table 7: Complete falsification protocol. Had SVN-Random maintained ≥60% at N=306, F-2 would be FALSIFIED.

Gate	Pre-registered Condition	Outcome	Status
F-1	SVN > VN on contrast ($p<0.05$)	+45.9%, $t=7.77$, $p<0.01$	✅ PASS
F-2	Fiedler > Random at N=306	10/10 vs 0/10, $t=17.68$	✅ PASS
F-3	SVN=VN when no mean collision	100% = 100% (GLOBAL task)	✅ PASS
F-4	Brain parcellation-SVN > global	Awaiting fMRI data	🟨 OPEN

F-5	LRGB Peptides: SVN > VN per-class	PyG: SVN > VN, p = 0.023 on matched SBM. Real LRGB pending.	🟧 PARTIAL
F-6	Stage-4 CMB: $\Delta N_{eff} \in [0.05, 0.30]$	Awaiting Stage-4-class CMB experiment	🟨 OPEN
F-7	Anti-numerology: $p<0.05$ for (2,3,6)	MC: $p=0.028 < 0.05$	✅ PASS

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Version History

v1.0 (March 2026): Initial public release. (Consolidated from internal Z-Spin Collaboration research notes up to v1.3.0.) Mean Collision Theorem, SVN architecture, NumPy experiments (§5.1–5.3), BBN D/H validation, application roadmap, FMO photosynthetic complex partial validation, production PyG validation (§5.4–5.6), Partition Quality Threshold (§5.5), effect size scaling (§5.6), Block Fiedler Mediation Theorem (§9.3). All internal references updated to v1.0 unified notation. 7 falsification gates documented (4 PASS, 2 OPEN, 1 PARTIAL). Zero free parameters.